SIMATS ENGINEERING

# SIMATS Online Programming Lab

C PROGRAMMING

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## CO2 - AT1 - LEVEL-1

Level Exam | </> C | 02 Sep 2026, 02:30 PM

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**Result Available**
Your score: **95 / 100**

QUESTIONS

LEVEL 1 — EASY

- Q1 Number
- Q2 Number
- Q3 Number
- Q4 Pattern

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Level 1 — Easy **divisors (E)** — Number ✓ Submitted

```
Problem: Write a program to count the number of divisors of a number entered by the user.

Sample Input:
Enter a number: 12

Sample Output:
Number of divisors: 6

Test Cases:
===TEST CASE 1===
Input:
Enter a number: 12
Expected Output:
Number of divisors: 6

===TEST CASE 2===
Input:
Enter a number: 12
Expected Output:
Number of divisors: 6
```

Your Code

```c
#include<stdio.h>
int main()
{
    int n,i,count=0;
    printf("Enter a number:");
    scanf("%d",&n);
    for(i=1;i<=n;i++)
    {
        if(n%i==0){
            count++;
        }
    }
    printf("\nNumber of divisors : %d",count);
   return 0;
}
```

Output

```
Enter a number:
Number of divisors : 6
```

Next

Your Code

```c
#include<stdio.h>
int main()
{
    int n,temp, sum=0;
    printf("Enter a number:");
    scanf("%d", &n);
    temp=n;
    while(temp>0)
    {
      sum=sum+(temp%10);
      temp=temp/10;
    }
    if(n%sum==0)
    {
      printf("\n%d is a Harshad number",n);
    }
    else{
      printf("\n%d is not a Harshad numner",n);
    }
    return 0;
}
```

Output

```
Enter a number:
18 is a Harshad number
```

```
Expected Output:
Decimal equivalent: 7
```

Your Code

```c
#include<stdio.h>
int main()
{
    int binary, decimal=0, base=1, rem;
    printf("Enter a binary number :");
    scanf("%d",&binary);
    while(binary>0)
    {
      rem=binary%10;
      decimal=decimal+rem*base;
      base=base*2;
      binary=binary/10;
    }
    printf("\nDecimal equivalent : %d", decimal);
    return 0;
}
```

Output

```
Enter a binary number :
Decimal equivalent : 10
```

```
********
```

Your Code

```c
#include<stdio.h>
int main()
{
    int n,i,j;
    printf("The pattern of the pyramid\n");
    scanf("%d", &n);
    for(i=1;i<=n;i++){
        for(j=1;j<=n-i;j++){
            printf(" ");
        }
        for(j=1;j<=2*i-1;j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Output

```
The pattern of the pyramid
   *
  ***
 *****
*******
```